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Law of Large Number

1.1 Independence

Definition 1.1.1 ► Independence

- Two events A and B are **independent** if $P(A \cap B) = P(A)P(B)$.
- Two random variables X and Y are **independent** if for all $C, D \in \mathcal{R}$,

$$P(X \in C, Y \in D) = P(X \in C)P(Y \in D)$$

that is, the events $A = \{X \in C\}$ and $B = \{Y \in D\}$ are independent.

- Two σ -fields $\mathcal{F}$ and $\mathcal{G}$ are **independent** if for all $A \in \mathcal{F}$ and $B \in \mathcal{G}$ the events A and B are independent.

Exercise 1.1.2

1. Show that if A and B are independent, then so are A^c and B , A and B^c , and A^c and B^c .
2. Conclude that events A and B are independent if and only if their indicator random variables 1_A and 1_B are independent.

Proof. 1. If A and B are independent, then

$$P(A \cap B) = P(A)P(B)$$

Note that

$$P(A \cap B) + P(A^c \cap B) = P(B)$$

We have

$$P(A^c \cap B) = P(B)(1 - P(A)) = P(B)P(A^c)$$

Similarly,

$$P(A \cap B^c) = P(A)P(B^c)$$

$$P(A^c \cap B^c) = P(B^c) - P(A \cap B^c) = P(B^c) - P(A)P(B^c) = P(A^c)P(B^c)$$

2.

$$\sigma(1_A) = \{\emptyset, A, A^c, \Omega\}, \quad \sigma(1_B) = \{\emptyset, B, B^c, \Omega\}$$

If $\sigma(1_A)$ and $\sigma(1_B)$ are independent, it is clear that A and B are as well. Conversely, only need to show that $\emptyset$ and A are independent, the remaining follows from 1. and symmetry. We have

$$P(\emptyset \cap A) = 0 = P(\emptyset)P(A)$$

□

Definition 1.1.3

- σ -fields $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n$ are **independent** if whenever $A_i \in \mathcal{F}_i$ for $i = 1, \dots, n$, we have

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i)$$

- Random variables $X_1, \dots, X_n$ are **independent** if whenever $B_i \in \mathcal{R}$ for $i = 1, \dots, n$ we have

$$P\left(\bigcap_{i=1}^n \{X_i \in B_i\}\right) = \prod_{i=1}^n P(X_i \in B_i)$$

- Sets $A_1, \dots, A_n$ are **independent** if whenever $I \subset \{1, \dots, n\}$ we have

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

Remark.

The last definition looks slightly different from the former two, but it is consistent with the independence of the indicator variables 1_{A_i} , $1 \leq i \leq n$.

Exercise 1.1.4

Let $A_1, A_2, \dots, A_n$ be independent. Show

1. $A_1^c, A_2, \dots, A_n$ are independent;
2. $1_{A_1}, \dots, 1_{A_n}$ are independent.

Proof. 1. If $1 \notin I$, then it is obvious that

$$P(\cap_{i \in I} A_i) = \prod_{i \in I} P(A_i).$$

If $1 \in I$, note that

$$\begin{aligned} P(\cap_{i \in I \setminus \{1\}} A_i \cap A_1^c) &= P(\cap_{i \in I \setminus \{1\}} A_i) - P(\cap_{i \in I \setminus \{1\}} A_i \cap A_1) \\ &= \prod_{i \in I \setminus \{1\}} P(A_i) - \prod_{i \in I} P(A_i) \\ &= (1 - P(A_1)) \prod_{i \in I \setminus \{1\}} P(A_i) \\ &= P(A_1^c) \prod_{i \in I \setminus \{1\}} P(A_i) \end{aligned}$$

2.

$$\sigma(1_{A_i}) = \{\emptyset, A_i^c, A_i, \Omega\}$$

Take $B_i \in \sigma(1_{A_i})$, $i = 1, \dots, n$. $B_i = \emptyset$ for some i , then both $P(\cap_{i=1}^n B_i)$ and $\prod_{i=1}^n P(B_i)$ are zero. The remainings follows from 1.

□

Example 1.1.5

Let X_1, X_2, X_3 be independent random variables with

$$P(X_i = 0) = P(X_i = 1) = 1/2$$

Let $A_1 = \{X_2 = X_3\}$, $A_2 = \{X_3 = X_1\}$ and $A_3 = \{X_1 = X_2\}$. These events are pairwise independent since if $i \neq j$, then

$$P(A_i \cap A_j) = P(X_1 = X_2 = X_3) = 1/4 = P(A_i)P(A_j)$$

but they are not independent since

$$P(A_1 \cap A_2 \cap A_3) = 1/4 \neq 1/8 = P(A_1)P(A_2)P(A_3)$$

1.1.1 Sufficient Conditions

Definition 1.1.6 ▶ Independence between collections

Collections of sets $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n \subset \mathcal{F}$ are said to be **independent** if whenever $A_i \in \mathcal{A}_i$ and $I \subset \{1, \dots, n\}$ we have $P(\bigcap_{i \in I} A_i) = \prod_{i \in I} P(A_i)$.

Lemma 1.1.7

Without loss of generality we can suppose each $\mathcal{A}_i$ contains Ω . In this case the condition is equivalent to

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i) \text{ whenever } A_i \in \mathcal{A}_i$$

since we can set $A_i = \Omega$ for $i \notin I$.

Definition 1.1.8 ▶ π, λ -systems

- We say that $\mathcal{A}$ is a **π -system** if it is closed under intersection, that is, if $A, B \in \mathcal{A}$ then $A \cap B \in \mathcal{A}$.
- We say that $\mathcal{L}$ is a **λ -system** if
 - (i) $\Omega \in \mathcal{L}$.
 - (ii) If $A, B \in \mathcal{L}$ and $A \subset B$, then $B - A \in \mathcal{L}$.
 - (iii) If $A_n \in \mathcal{L}$ and $A_n \uparrow A$, then $A \in \mathcal{L}$.

Theorem 1.1.9 ▶ $\pi - \lambda$ Theorem

If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system that contains $\mathcal{P}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Theorem 1.1.10

Suppose $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent and each $\mathcal{A}_i$ is a π -system. Then $\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \dots, \sigma(\mathcal{A}_n)$ are independent.

Proof sketch.

对于任一取定的事件 F , “(取定任一集族中) 与 F 独立” 的性质集是一个 λ -system. 由于 $\mathcal{A}_1$ 是 π -system, 于是 $\mathcal{A}_1$ 中与 F 独立的集合包含了 $\sigma(\mathcal{A}_1)$, 通过任意的 F , 传递归纳集族的独立性一步步“升级”到它们的生成 σ -代数上.

Remark.

It is not easy to show that if $A, B \in \mathcal{L}$ then $A \cap B \in \mathcal{L}$ or $A \cup B \in \mathcal{L}$, but it is easy to check that if $A, B \in \mathcal{L}$ with $A \subseteq B$, then $B - A \in \mathcal{L}$.

Proof. Let $F \in \mathcal{P}$ be any event. We define

$$\mathcal{L} := \{A \in \mathcal{A}_1 : P(A \cap F) = P(A)P(F)\}$$

We check that $\mathcal{L}$ is a λ -system. (i) follows from $P(A \cap \Omega) = P(A) \cdot 1 = P(A)P(\Omega)$. To show (ii), we take $A, B \in \mathcal{L}, B \supseteq A$. Note that

$$(B - A) \cap F = (B \cap F) - (A \cap F),$$

we have

$$\begin{aligned} P((B \setminus A) \cap F) &= P((B \cap F) - (A \cap F)) \\ &= P(B \cap F) - P(A \cap F) \\ &= P(B)P(F) - P(A)P(F) \\ &= (P(B) - P(A))P(F) = P((B - A))P(F) \end{aligned}$$

which shows that $(B - A) \in \mathcal{L}$

For (iii), take $B_n \in \mathcal{L}, B_n \uparrow B$. Then since $(B_n \cap F) \uparrow B \cap F$, we have

$$\begin{aligned} P(B \cap F) &= \lim_{n \rightarrow \infty} P(B_n \cap F) \\ &= \lim_{n \rightarrow \infty} P(B_n)P(F), \\ &= P(B)P(F) \end{aligned}$$

which implies that $B \in \mathcal{L}$. Thus $\mathcal{L}$ is a λ -system. It follows from $\pi - \lambda$ Theorem that $\mathcal{L} \supseteq \sigma(\mathcal{A}_1)$ For each $A_2, \dots, A_n, A_i \in \mathcal{A}_i, i = 2, 3, \dots, n$, by taking $F = \bigcap_{i=2}^n A_i$, we have

$$P(A \cap A_2 \cap \dots \cap A_n) = P(A)P\left(\bigcap_{i=2}^n A_i\right) = P(A)P(A_2) \dots P(A_n), \quad \forall A \in \sigma(\mathcal{A}_1)$$

We get that

if $\mathcal{A}_1, \dots, \mathcal{A}_n$ are independent π -systems, then $\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$ are as well (*)

By applying (*) into $\mathcal{A}_2, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$, we get $\sigma(\mathcal{A}_2), \mathcal{A}_3, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$ are

independent. The proof completes after n iterations. $\square$

Theorem 1.1.11

In order for $X_1, \dots, X_n$ to be independent, it is sufficient that for all $x_1, \dots, x_n \in (-\infty, \infty]$

$$P(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$$

Proof. Let $\mathcal{A}_i$ be the collection of the sets of the form $\{X_i \leq x_i\}$. Since

$$\{X_i \leq x\} \cap \{X_i \leq y\} = \{X_i \leq x \wedge y\}$$

$\mathcal{A}_i$ is a π -system. Since we allowed $x_i = \infty$, $\Omega \in \mathcal{A}_i$. Exercise ?? implies that $\sigma(\mathcal{A}_i) = \sigma(X_i)$, so the result follows from Theorem 1.1.11. $\square$

Theorem 1.1.12

Suppose $\mathcal{F}_{i,j}$, $1 \leq i \leq n$, $1 \leq j \leq m(i)$ are independent π -systems and let $\mathcal{G}_i = \sigma(\bigcup_j \mathcal{F}_{i,j})$. Then $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent.

Note.

对于独立的 π -systems, 可以把它们按任意分组方式, “攢起来” 做 σ -代数扩张, 得到新的独立系统.

Proof. Let $\mathcal{A}_i$ be the collection of sets of the form $\bigcap_j A_{i,j}$ where $A_{i,j} \in \mathcal{F}_{i,j}$. $\mathcal{A}_i$ is a π -system that contains Ω and contains $\bigcup_j \mathcal{F}_{i,j}$, so Theorem 1.1.10 implies $\sigma(\mathcal{A}_i) = \mathcal{G}_i$ are independent. $\square$

Theorem 1.1.13

If for $1 \leq i \leq n$, $1 \leq j \leq m(i)$, $X_{i,j}$ are independent and $f_i : \mathbf{R}^{m(i)} \rightarrow \mathbf{R}$ are measurable, then $f_i(X_{i,1}, \dots, X_{i,m(i)})$ are independent.

Proof. $\sigma(X_{i,j})$, $1 \leq i \leq n$, $1 \leq j \leq m(i)$ are independent π -systems. Then

$\mathcal{G}_i = \sigma\left(\bigcup_j (\sigma(X_{i,j}))\right)$ are independent. Take any Borel set $B \subseteq \mathbb{R}$, then

$$\begin{aligned} (f_i(X_{i,1}, \dots, X_{i,m(i)}))^{-1}(B) &= (X_{i,1}, \dots, X_{i,m(i)})^{-1}(f_i^{-1}(B)) \\ &\subseteq \sigma\left(\bigcup_j (\sigma(X_{i,j}))\right) \end{aligned}$$

Thus $f_i(X_{i,1}, \dots, X_{i,m(i)})$ are independent. □

1.1.2 Independence, Distribution, and Expectation

Theorem 1.1.14

Suppose $X_1, \dots, X_n$ are independent random variables and X_i has distribution μ_i . Then $(X_1, \dots, X_n)$ has distribution $\mu_1 \times \dots \times \mu_n$.

Proof.

$$\begin{aligned} P((X_1, \dots, X_n) \in A_1 \times \dots \times A_n) &= P(X_1 \in A_1, \dots, X_n \in A_n) \\ &= \prod_{i=1}^n P(X_i \in A_i) \\ &= \prod_{i=1}^n \mu_i(A_i) \\ &= (\mu_1 \times \dots \times \mu_n)(A_1 \times \dots \times A_n) \end{aligned}$$

The above shows that the distribution of $(X_1, \dots, X_n)$ and the measure $\mu_1 \times \dots \times \mu_n$ agree on the sets of the form $A_1 \times \dots \times A_n$, which forms a π -system generates $\mathcal{R}^n$. Since the collection of the sets on which two measures here agree forms a $\mathcal{L}$ -system. Then it follows from Theorem 1.1.9 that $\mu_1 \times \dots \times \mu_n$ and $(X_1, \dots, X_n)$ agree on $\mathcal{R}^n$. □

Theorem 1.1.15

Suppose X and Y are independent and have distributions μ and ν . If $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a measurable function with $h \geq 0$ or $E|h(X, Y)| < \infty$, then

$$Eh(X, Y) = \iint h(x, y)\mu(dx)\nu(dy)$$

In particular, if $h(x, y) = f(x)g(y)$ where $f, g : \mathbf{R} \rightarrow \mathbf{R}$ are measurable functions with $f, g \geq 0$ or $E|f(X)|$ and $E|g(Y)| < \infty$, then

$$Ef(X)g(Y) = Ef(X) \cdot Eg(Y)$$

Proof. From the Changing of Variables and the Fubini's Theorem, we have

$$Eh(X, Y) = \int h \, d(\mu \times \nu) = \iint h(x, y) \mu(dx) \nu(dy)$$

To show the second part, we first assume that $f, g \geq 0$, then

$$\begin{aligned} Ef(X)g(Y) &= \iint f(x)g(y) \mu(dx) \nu(dy) \\ &= \int g(y) \int f(x) \mu(dx) \nu(dy) \\ &= \int g(y) Ef(X) \nu(dy) \\ &= Ef(X) Eg(Y) \end{aligned}$$

For another assumption, we apply the above result to the nonnegative function $|f|, |g|$, to get $E|h(X, Y)| < \infty$, and we repeat the last argument to prove the desired result. $\square$

Theorem 1.1.16

If $X_1, \dots, X_n$ are independent and have

- (a) $X_i \geq 0$ for all i , or
- (b) $E|X_i| < \infty$ for all i , then

$$E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n EX_i$$

that is, the expectation on the left exists and has the value given on the right.

1.1.3 Sums of Independent Random Variables

Theorem 1.1.17

If X and Y are independent, $F(x) = P(X \leq x)$, and $G(y) = P(Y \leq y)$, then

$$P(X + Y \leq z) = \int F(z - y) dG(y)$$

The integral on the right-hand side is called the **convolution** of F and G and is denoted $F * G(z)$. The meaning of $dG(y)$ will be explained in the proof.

Proof.

$$\begin{aligned} P(X + Y \leq z) &= \int 1_{(x+y \leq z)} d(\mu \times \nu) \\ &= \int \int 1_{(x+y \leq z)} d\mu d\nu \\ &= \int F(z - y) d\nu \\ &= \int F(z - y) dG(y) \end{aligned}$$

where $dG(y)$ is just a notation for $d\nu$. We apply this notation motivated from the Lebesgue-Stieljes integration. $\square$

Theorem 1.1.18

Suppose that X with density f and Y with distribution function G are independent. Then $X + Y$ has density

$$h(x) = \int f(x - y) dG(y)$$

When Y has density g , the last formula can be written as

$$h(x) = \int f(x - y) g(y) dy$$

Proof.

$$F(x) = \int_{-\infty}^x f(x) dx$$

Since everything is nonnegative, from Fubini's theorem, we get

$$\begin{aligned} P(X + Y \leq z) &= \int \int_{-\infty}^{z-y} f(x) dx dG(y) \\ &= \int \int_{-\infty}^z f(x-y) dx dG(y) \\ &= \int_{-\infty}^z \int f(x-y) dG(y) dx \\ &= \int_{-\infty}^z h(x) dx \end{aligned}$$

which shows that $X + Y$ has density $h(x)$.

When Y has density g , then

$$\int f(x-y) dG(y) = \int f(x-y) g(y) dy$$

□

Example 1.1.19 ► The gamma density

The **gamma density** with parameters α and λ is given by

$$f(x) = \begin{cases} \lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha) & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$.

Theorem 1.1.20

If $X = \text{gamma}(\alpha, \lambda)$ and $Y = \text{gamma}(\beta, \lambda)$ are independent, then $X + Y$ is $\text{gamma}(\alpha + \beta, \lambda)$. Consequently if $X_1, \dots, X_n$ are independent exponential (λ) r.v's, then $X_1 + \dots + X_n$ has a $\text{gamma}(n, \lambda)$ distribution.

Example 1.1.21 ▶ Normal distribution

In Example 1.6.2, we introduced the normal density with mean μ and variance a ,

$$(2\pi a)^{-1/2} \exp(-(x - \mu)^2/2a).$$

Theorem 1.1.22

If $X = \text{normal}(\mu, a)$ and $Y = \text{normal}(\nu, b)$ are independent, then $X + Y = \text{normal}(\mu + \nu, a + b)$.

1.2 Weak Laws of Large Numbers